# PROJECT DOCUMENTATION

# STREAMING TICKET ASSIGNMENT FOR EFFICIENT SUPPORT OPERATIONS


**Submitted By**

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# 1.ABSTRACT

This project focuses on improving the efficiency of support operations by automating the assignment of support tickets using the ServiceNow platform. In large organizations, support teams often handle a high volume of tickets related to different services such as platform access issues, login problems, certificate requests, system errors, and other operational concerns. When these tickets are assigned manually, it can lead to delays in issue resolution, incorrect routing of tickets, increased workload for support staff, and reduced customer satisfaction.

To address these challenges, the project proposes an automated ticket routing system that intelligently assigns support tickets to the appropriate support groups based on the type of issue reported. The system is implemented using various features of the ServiceNow platform, including Users, Groups, Roles, Custom Tables, Access Control Lists (ACL), and Flow Designer. These components work together to create a structured environment where tickets are automatically analyzed and routed to the correct support team without manual intervention.

The project involves creating dedicated support groups for handling different categories of issues, such as platform-related problems and certificate-related requests. Automated workflows are then designed using Flow Designer to monitor newly created or updated records in the system. Based on predefined conditions, the workflows automatically assign tickets to the relevant support groups, ensuring faster and more accurate ticket distribution.

By implementing this automated approach, the organization can significantly reduce manual effort, improve response time, and ensure that issues are handled by the appropriate teams. The solution enhances operational efficiency, optimizes resource utilization, and improves the overall customer support experience by ensuring timely and effective resolution of support requests.

# 2.INTRODUCTION

## 2.1 Overview of ServiceNow

ServiceNow is a cloud-based platform that helps organizations automate business processes and manage IT services efficiently. It provides a centralized system for handling incidents, service requests, problems, and other support operations. ServiceNow is widely used in IT Service Management (ITSM) because it helps organizations track issues, manage workflows, and improve service delivery across different departments.ServiceNow operates using a table-based architecture, where all information such as users, incidents, groups, and tasks are stored as records in database tables. These tables are connected through reference fields, allowing the system to maintain relationships between different types of data. For example, a support ticket can be linked to a specific user or support group responsible for resolving the issue.

The platform also offers powerful automation tools such as Flow Designer, which allows organizations to create workflows that automatically perform actions when certain conditions are met. This helps reduce manual effort, improves efficiency, and ensures that tasks such as ticket assignment, notifications, and approvals are handled quickly and accurately. Overall, ServiceNow helps organizations streamline operations, improve productivity, and deliver better support services.

## 2.2 Importance of Automated Ticket Assignment

In organizations that offer multiple products and services, customer support teams receive a large number of support tickets every day. These tickets may include issues such as login problems, system errors, access requests, or service-related queries. When tickets are assigned manually, support staff must review each request and determine the appropriate team responsible for resolving the issue. This process can be time-consuming and may lead to several operational challenges such as delays in ticket resolution, incorrect ticket assignment, increased workload for support staff, and reduced customer satisfaction.

Automated ticket assignment helps overcome these challenges by using predefined rules and conditions to automatically route tickets to the correct support group. By analyzing the issue type or category of the ticket, the system immediately assigns it to the appropriate team without manual intervention. This ensures that the ticket reaches the right department quickly, allowing support teams to begin resolving the issue faster. As a result, automation improves response time, reduces human errors, enhances operational efficiency, and provides a better support experience for customers.

# 3. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To automate the process of ticket assignment in ServiceNow.
- To create users, groups, and roles for managing support teams.
- To design a custom table for handling operational issues.
- To configure access control for secure data management.
- To implement automated flows for routing tickets to appropriate groups.
- To improve operational efficiency and customer satisfaction.
- To reduce manual effort in ticket management.

# 4. TOOLS AND TECHNOLOGIES USED

| **Tool / Technology** | **Purpose** |
|---|---|
| ServiceNow Platform | Used to build and automate the ticket management system |
| Flow Designer | Used to automate ticket assignment workflows |
| Users and Groups | Used to manage support team members |
| Roles | Used to control system access |
| Access Control Lists (ACL) | Used to secure system data |
| Custom Tables | Used to store ticket-related information |

# 5. SYSTEM ARCHITECTURE /WORKFLOW

User Request / Ticket Creation → Operations Related Table → Flow Designer Trigger → Issue Classification → Automatic Group Assignment → Ticket Resolution.

| Step | Stage | Component | Key Activity & Impact |
|---|---|---|---|
| **1. Ticket Creation & Storage** | Stage 1 & 2 | User / Support Ticket Operations Related Table | **Activity:** A user submits a ticket describing an issue (e.g., login failure). This ticket is stored in a custom ServiceNow table<br>**Impact:** Captures critical data and serves as the initial system input. |
| **2. Automated Classification & Assignment** | Stage 3, 4, & 5 | Flow Designer Trigger Issue Classification Automatic Ticket Assignment | **Activity:** A Flow Designer workflow triggers, classifies the issue category (e.g., 404 error), and automatically assigns the ticket to the appropriate group (e.g., Platform Support).<br>**Impact:** Ensures immediate categorization and speeds up the time to the correct technician. |
| **3. Resolution & Closure** | Stage 6 | Support Team Resolution | **Activity:** The assigned support group reviews the ticket, resolves the issue, and updates the ticket status<br>**Impact:** Leads to resolution, a complete audit trail, and faster overall support operation efficiency. |

## Workflow Explanation

Milestone 1 — Ticket Creation and Storage: Users create support tickets and the details are stored in the Operations Related table in ServiceNow

Milestone 2 — Automated Ticket Routing: Flow Designer automatically analyzes the issue type and assigns the ticket to the appropriate support group such as Platform

# 6. PROJECT IMPLEMENTATION STEPS

The implementation of the project was carried out in a structured sequence of steps as described below:

## Step 1: Creating Users

The first step in the implementation process involved creating users in the ServiceNow platform. Users represent individuals who interact with the system, such as support staff responsible for resolving tickets. The required user details such as name, email address, and role were entered while creating the user records. These users were later assigned to specific support groups that handle different types of issues.

## Step 2: Creating Support Groups

After creating users, support groups were created to represent teams responsible for resolving different types of tickets. For this project, two groups were created: Platform Support Group and Certificates Support Group. Each group consists of users who are responsible for resolving issues related to their specific area. These groups help organize support staff and simplify the ticket assignment process.

## Step 3: Creating Roles

Roles were created in the system to define access permissions for users. Roles ensure that only authorized users can access or modify certain records in the system. In this project, roles such as Platform Role and Certificate Role were created. These roles were later assigned to the respective users and groups to maintain proper security and access control within the system.

## Step 4: Creating a Custom Table

A custom table named Operations Related was created in the ServiceNow platform to store support ticket information. This table acts as the primary location where all support requests are recorded. Fields such as Issue Type, Description, and Assigned Group were created in this table. The Issue Type field contains options such as login issues, platform errors, certificate requests, and user expiration issues.

## Step 5: Assigning Users and Roles to Groups

Once the users, groups, and roles were created, users were assigned to the appropriate groups and roles. For example, users responsible for handling certificate-related issues were assigned to the Certificates Group, while users responsible for platform-related issues were assigned to the Platform Group. This step ensured that each support group had the necessary members to resolve assigned tickets.

## Step 6: Configuring Access Control Lists (ACL)

Access Control Lists (ACLs) were configured to manage security and control access to the custom table. ACL rules were created to ensure that only authorized users with specific roles could read, create, update, or delete records in the Operations Related table. This helped protect sensitive data and maintain proper system security.

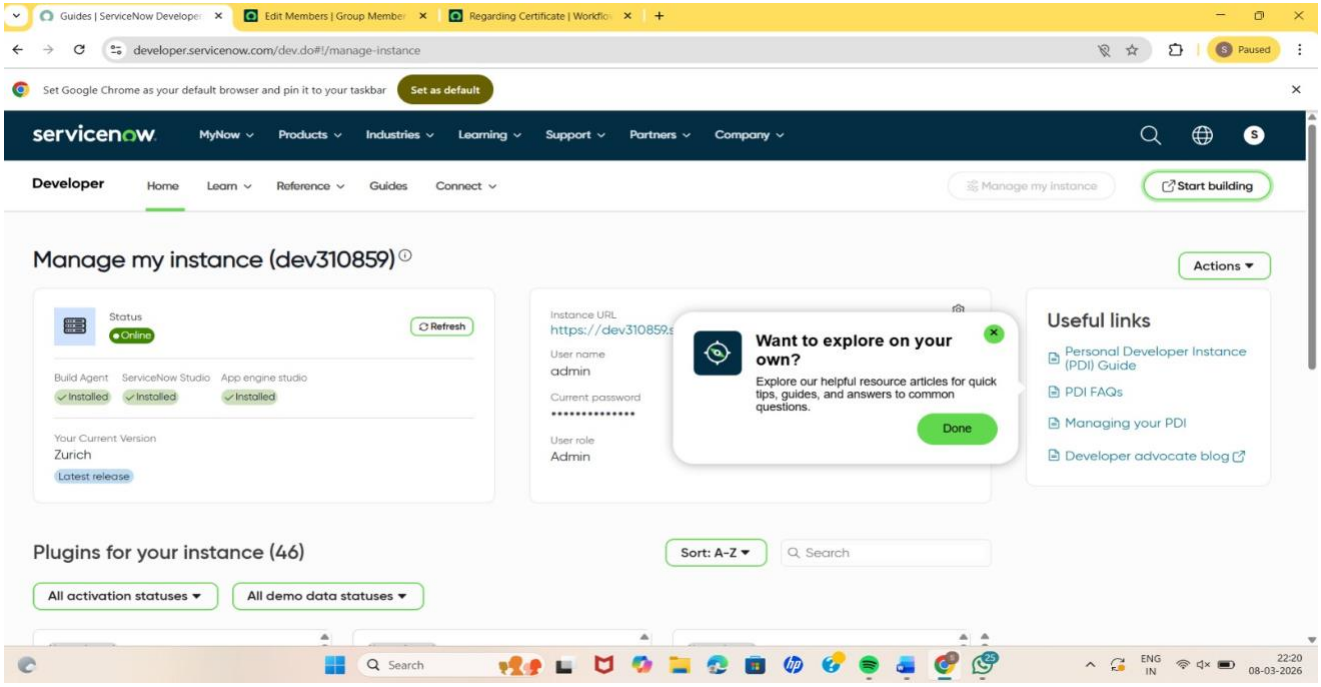
## Step 7: Creating Automated Flows using Flow Designer

Automation was implemented using Flow Designer, which allows workflows to be created through a visual interface. Two flows were created in this project. The first flow handles certificate-related issues, while the second flow handles platform-related issues. These flows are triggered whenever a new ticket is created in the Operations Related table.

## Step 8: Running the Automated Ticket Assignment Process

When a ticket is created, the Flow Designer workflow automatically checks the Issue Type field. Based on the selected issue category, the system assigns the ticket to the appropriate support group. For example, if the issue is related to certificates, the ticket is assigned to the Certificates Group, and if the issue is related to platform errors or login problems, the ticket is assigned to the Platform Group. This automated process ensures accurate ticket routing and improves support efficiency.

# 7.SCREENSHOTS



Guides | ServiceNow Developer | Groups | ServiceNow | Regarding Certificate | Workflow

dev310859.service-now.com/now/nav/ui/classic/params/target/sys_user_group_list.do%3Fsysparm_query%3DGOTOname%253E%253Dplatform%26sysparm_view%26sysparm_clear_st...

Set Google Chrome as your default browser and pin it to your taskbar | Set as default

servicenow | All | Favorites | History | Workspaces | Admin | Groups | Search

Groups | Name | Search | Actions on selected rows... | New

All > Name >= platform

| Name | Description | Active | Manager | Parent | Updated |
|---|---|---|---|---|---|
| platform | | true | (empty) | (empty) | 2026-03-08 07:27:35 |
| Problem Analyzers | Group that works on analyzing and fixing... | true | (empty) | (empty) | 2026-01-04 21:11:33 |
| Problem Solving | Group that works on analyzing and fixing... | true | (empty) | (empty) | 2026-01-04 21:10:15 |
| Procurement | Responsible for ordering catalog items, ... | true | (empty) | (empty) | 2021-06-11 18:50:16 |
| Project Mgmt | Responsible for Return Material Authorization approvals and RMA number allocations in Inventory Management | true | (empty) | (empty) | 2024-05-09 18:42:19 |
| Recommendation Admin | | true | (empty) | (empty) | 2019-02-14 16:39:04 |
| Report Access Request Approvers | | true | (empty) | (empty) | 2021-03-15 11:50:37 |
| RMA Approvers | Responsible for Return Material Authoriz... | true | (empty) | (empty) | 2026-01-05 13:12:50 |
| Service Desk | | true | Beth Anglin | (empty) | 2026-01-04 16:14:56 |
| Software | IT department responsible for all softwa... | true | (empty) | (empty) | 2021-06-11 18:53:27 |
| Survey Creators | This is a group of users who have survey... | true | (empty) | (empty) | 2019-02-11 11:36:56 |
| Team Development Code Reviewers | Review, approve and/or reject the code p... | true | (empty) | (empty) | 2013-12-01 15:47:36 |
| US Presidents Group 1 | Demo user group for Password Reset Appli... | true | (empty) | (empty) | 2013-07-25 15:20:25 |
| US Presidents Group 2 | Demo user group for Password Reset Appli... | true | (empty) | (empty) | 2013-07-25 15:21:07 |

1 to 14 of 14

Search | ENG IN | 22:21 08-03-2026

Guides | ServiceNow Developer | Groups | ServiceNow | Regarding Certificate | Workflow

dev310859.service-now.com/now/nav/ui/classic/params/target/sys_user_group_list.do%3Fsysparm_query%3DGOTOname%253E%253Dcertificates%26sysparm_view%26sysparm_clear_...

Set Google Chrome as your default browser and pin it to your taskbar | Set as default

servicenow | All | Favorites | History | Workspaces | Admin | Groups | Search

Groups | Name | Search | Actions on selected rows... | New

All > Name >= certificates

| Name | Description | Active | Manager | Parent | Updated |
|---|---|---|---|---|---|
| certificates | | true | (empty) | (empty) | 2026-03-08 07:26:59 |
| Change Management | Change Management Group | true | (empty) | (empty) | 2015-06-09 22:35:12 |
| Conditional Script Writer | This is the default group introduced by ... | true | (empty) | (empty) | 2025-02-25 18:04:32 |
| Creator Studio Restricted Users | Provides access to the Creator Studio de... | true | (empty) | (empty) | 2023-12-07 15:58:22 |
| Creator Studio Users | Provides access to the Creator Studio de... | true | (empty) | (empty) | 2023-12-07 15:48:42 |
| Database | | true | Don Goodliffe | (empty) | 2023-03-13 19:02:18 |
| Database Atlanta | Manages databases hosted in Atlanta | true | (empty) | Database | 2024-03-30 11:47:18 |
| Database San Diego | Manages databases hosted in San Diego | true | (empty) | Database | 2024-03-30 11:46:37 |
| eCAB Approval | eCAB Approvers | true | (empty) | (empty) | 2011-09-30 01:46:44 |
| Field Services | | true | (empty) | (empty) | 2021-06-07 18:11:34 |
| Hardware | IT department responsible for all hardwa... | true | (empty) | (empty) | 2021-06-11 18:54:14 |
| Help Desk | | true | ATF Change Management | (empty) | 2026-01-12 01:50:32 |
| Incident Management | Incident Management Group | true | (empty) | (empty) | 2017-05-16 04:13:49 |
| IT Finance CAB | | true | (empty) | (empty) | 2021-10-30 14:00:17 |

1 to 20 of 40

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Guides | ServiceNow Developer | Edit Members | Group Role | Se | Regarding Certificate | Workflo

dev310859.service-now.com/now/nav/ui/classic/params/target/sys_m2m_template.do%3Fsys_is_list%3Dtrue%26sys_is_related_list%3Dtrue%26sys_target%3Dsys_group_has_role%26sy...

Set Google Chrome as your default browser and pin it to your taskbar **Set as default**

servicenow All Favorites History Workspaces **Group Role - Edit Members** Search

Edit Members **Cancel** **Save**

**Add Filter** **Run filter**

-- choose field -- | -- oper -- | -- value --

Collection

Roles List

certificates

access_analyzer_admin
access_analyzer_ai_user
action_category_creator
action_designer
action_write_enabled
activity_admin
activity_creator
actsub_admin
actsub_user
admin
agent_admin
agent_security_admin
agent_workspace_user
ai_native_experience_analytics_admin
ai_native_experience_analytics_viewer

Certification_role

**Cancel** **Save**

Search ENG IN 22:22 08-03-2026

Guides | ServiceNow Developer | action_designer | Role | Service | Regarding Certificate | Workflo

dev310859.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3D11bc76b20b0222008e650851c5673a66%26sysparm_record_target%3Dsys_user_role

Set Google Chrome as your default browser and pin it to your taskbar **Set as default**

servicenow All Favorites History Workspaces Admin **Role - action_designer** Search

Role
action_designer

**Update** **Delet**

Name: action_designer

Application: Global

Elevated privilege: ☐

Description: action designer role enables users to launch Action Designer

**Update** **Delete**

Contains Roles (2) | Applications with Role (1) | Modules with Role (8) | Custom Tables

for text | Search | Actions on selected rows... | **New** | **Edit...**

Role = action_designer

| Contains |
| --- |
| action_write_enabled |
| sn_workflow_studio.workflow_studio_read |

1 to 2 of 2

Search ENG IN 22:2 08-03-202

Guides | ServiceNow Developer | certificates | Group | ServiceNo | Regarding Certificate | Workflo

dev310859.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysId%3D659e4b70c3277210ea57befdd401310b%26

Set Google Chrome as your default browser and pin it to your taskbar | Set as default

Workflow Studio | Regarding Certificate Flow • Global | Regarding Platform Flow • Global

**Regarding Certificate** Active

View: | Test | Edit flow | Deactivate

TRIGGER

Operations related Created or Updated where (Issue is regarding certificates)

ACTIONS

1 Update Operations related Record

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data Expand All

- Flow Variables
- Trigger - Record Created or Updated
- 1 - Update Record

Read-only | Status: Published | Application: Global

Search | ENG IN | 22:27 08-03-2026

Search | ENG IN | 22:23 08-03-2026

Guides | ServiceNow Developer | certificates | Group | ServiceNo | Regarding Platform | Workflow

dev310859.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysId%3D03929b38c3277210ea57befdd40131e5%26

Set Google Chrome as your default browser and pin it to your taskbar | Set as default

Workflow Studio | Regarding Certificate Flow • Global | Regarding Platform Flow • Global

**Regarding Platform** Active

View: | Test | Edit flow | Deactivate

TRIGGER

Operations related Created or Updated where (Issue is unable to login to platform; Issue is 404 error; Issue is regarding user expired)

ACTIONS

1 Update Operations related Record

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data Expand All

- Flow Variables
- Trigger - Record Created or Updated
- 1 - Update Record

Read-only | Status: Published | Application: Global

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# 8. RESULTS AND OUTPUT

After implementing the automated ticket assignment system in the ServiceNow platform, several positive outcomes were observed. The system successfully improved the efficiency of the support process and ensured that tickets were handled in a more organized and faster manner. The following results were achieved:

- Tickets were automatically assigned to the correct support groups based on the issue type.
- Manual ticket routing by administrators or support managers was eliminated.
- Support teams received tickets immediately after they were created in the system.
- The chances of incorrect ticket assignment were significantly reduced.
- Issue resolution time improved because tickets reached the right team without delay.
- The overall workflow of the support process became more structured and efficient.
- Support staff were able to focus more on resolving issues instead of managing ticket distribution.
- The system ensured better coordination between different support teams.
- The automated process reduced human errors that commonly occur during manual assignment.
- Ticket tracking and management became easier through the centralized ServiceNow platform.
- The workload among support teams became more balanced due to proper routing of tickets.
- Customer satisfaction increased because issues were addressed more quickly and accurately.
- The automation improved the overall operational efficiency of the support department.
- The system demonstrated how workflow automation can enhance service management processes in organizations.

# 9. ADVANTAGES

The implementation of the automated ticket assignment system provides several benefits to the support operations. These advantages help improve efficiency, accuracy, and overall service quality within the organization.

Faster Ticket Assignment:

The system automatically assigns tickets to the appropriate support group as soon as they are created. This eliminates delays that occur in manual routing and ensures that issues are addressed quickly.

Reduced Manual Effort:

Automation removes the need for support managers or administrators to manually review and assign tickets. This reduces the workload on staff and allows them to focus on resolving issues rather than managing ticket distribution.

Improved Customer Satisfaction:

Since tickets are routed instantly to the correct support team, customer issues are resolved more quickly. Faster response and resolution times lead to a better support experience for customers.

Efficient Resource Utilization:

The system ensures that tickets are assigned to the correct teams based on the type of issue. This helps utilize the skills and expertise of support staff effectively and prevents unnecessary workload on unrelated teams.

Better Ticket Tracking:

All tickets are stored and managed within the ServiceNow platform, allowing easy tracking of ticket status, assignment, and resolution progress. This improves transparency and accountability in support operations.

Increased Productivity:

. By reducing manual tasks and improving workflow efficiency, support teams can handle more tickets in less time. This increases the overall productivity of the support department and improves service performance.

# 10.CONCLUSION

The implementation of the automated ticket routing system significantly improved the efficiency of support operations at ABC Corporation. By using ServiceNow features such as users, groups, roles, Access Control Lists (ACL), and Flow Designer, the process of assigning support tickets to the appropriate teams was successfully automated. This reduced the need for manual ticket routing and ensured that each support request was directed to the correct team based on the issue type.

The automated system helped reduce delays in ticket assignment and minimized the chances of incorrect routing. As a result, support teams were able to receive tickets faster and begin resolving issues without unnecessary waiting time. This improved the overall workflow of the support process and allowed support staff to focus more on resolving problems rather than managing ticket distribution.

Another important outcome of the system was improved operational efficiency. With automation handling repetitive tasks, the workload on support administrators was reduced, and the process became more organized and reliable. The centralized system also made it easier to track, monitor, and manage support tickets within the organization.

Additionally, the faster response time and accurate ticket routing contributed to higher customer satisfaction. Users were able to receive quicker support for their issues, which improved their overall experience with the organization's services.

In conclusion, the project demonstrates how workflow automation using the ServiceNow platform can streamline support operations, reduce manual effort, and improve service quality. The automated ticket assignment system provides an effective solution for managing support requests and can be further enhanced in the future with advanced features to improve IT service management processes.

# 11. REFERENCES


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